

Solid State Transformer



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Compute Project

Solid State Transformer (SST) Specification

Medium Voltage to 800 VDC Power Conversion Platform

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Author: Google, Microsoft, Nvidia

Version History

Date	Version #	Author	Description
5/20/26	0.3	- Google: Jing Wang, Sangyoun “SY” Kim, Satarupa Bal, Kei Hao, Dragan Ristanovic, Ashish Upreti, Jason Adrian, Christina Peabody - Microsoft: Ehsan Nasr, Pritesh Patel, Xianyong Feng - Nvidia: Mike Tu	first draft of SST product for OCP community

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1. License

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2. Compliance with OCP Tenets

2.1.Openness

This specification aims to establish a common, open framework for Solid State Transformers (SSTs) as a power conversion element enabling future LVDC distribution architectures in data centers. By defining clear electrical, functional, and interface expectations at the medium voltage to DC conversion layer, this work supports interoperability between SSTs, downstream OCP aligned power racks, and IT infrastructure. The intent is to encourage multivendor innovation while reducing integration friction through transparent requirements, well defined boundaries, and alignment with open standards and OCP design principles.

2.2.Efficiency

As the focus of this specification, Solid State Transformers, enable higher efficiency power delivery by combining medium voltage power conversion and direct DC output. By integrating advanced power electronics, modular architectures, and optimized control, SSTs support high conversion efficiency across a wide operating range, contributing to reduced electrical losses and improved overall data center power efficiency.

2.3.Impact

Besides introducing breakthroughs in technology, standardizing SST functional and performance expectations has a significant impact on data center power architectures by enabling new levels of density, and integration. A common SST specification, formed through iterative efforts among key authors, starts from concept, matures over multiple revisions, enables a broader supplier ecosystem and encourages feedback for improvements. It accelerates adoption of DC centric architectures, and creates a clearer pathway for coupling energy with server racks that enable the next generation of intelligence infrastructure. This alignment can reduce bespoke designs, improve supply chain scalability, and enable the reuse of power infrastructure across multiple data center generations in a long term vision.

2.4.Scale

This specification supports scaling across multiple power levels and deployment sizes, from distributed or block redundant systems supporting multi-megawatts loads with predictable reliability and availability metrics.

2.5.Sustainability

This specification serves as the foundation for enabling higher density data center power distribution architectures through the integration of silicon power electronics semiconductors based products called Solid State Transformer. By transitioning away from traditional copper and iron transformer technologies, it removes long-standing material and design constraints, fostering a more agile and efficient power ecosystem. Furthermore, these advancements directly contribute to the evolution of the renewable energy industry, facilitating the integration of sustainable power sources and supporting the long-term vision of green data center infrastructure.

3. Scope

This document serves as the technical design specification for the Solid State Transformer (SST), which is one of the building blocks for the Direct Current power distribution system in the Data Center industry.

Within the scope of this specification, a Medium Voltage Solid-State Transformer (MVSST) is defined as a **megawatt-scale power electronic converter**, which when coupled with energy storage could function as a Medium Voltage Uninterruptible Power Supply (MV-UPS). The MVSST itself replaces conventional 50/60Hz line-frequency Medium Voltage transformer, rectifier units and energy storage (as required by architecture). Among all engineering solutions to fulfill high-efficiency AC-to-DC conversion, SST is the most strategically innovative approach to enable downstream Low-Voltage Direct Current (LVDC) distribution to the IT rack interface. Beyond basic transformation, the SST provides advanced grid-interactive capabilities such as programmable MVAC response compliant with evolving grid codes, controlled LVDC voltage regulation for power distribution and IT rack requirements, and high-density data center design to enable power-dense racks (hundreds KW to MW). By utilizing medium-to-high-frequency magnetic components/transformers (MFT/HFT) and Wide-Bandgap (WBG) semiconductors such as Silicon Carbide (SiC) or emerging Gallium Nitride (GaN) potential, SST targets to achieve superior power density and offers a platform for future innovations in power distribution system design.

This document defines the product using a "**black-box**" approach, with the focus on interfaces as application level guidance (e.g. electrical, mechanical, thermal, environmental, control, communications, and behavior expectation etc). Vendors shall take the specification as a product framework to fill with engineering solutions to hit the design target.

A few examples for application architectures for SST are illustrated in [Appendix A](#). The current scope of this specification is defined according to the dotted frame in the architecture graph referenced.

4. Base product specifications

4.1. Product internal compositions

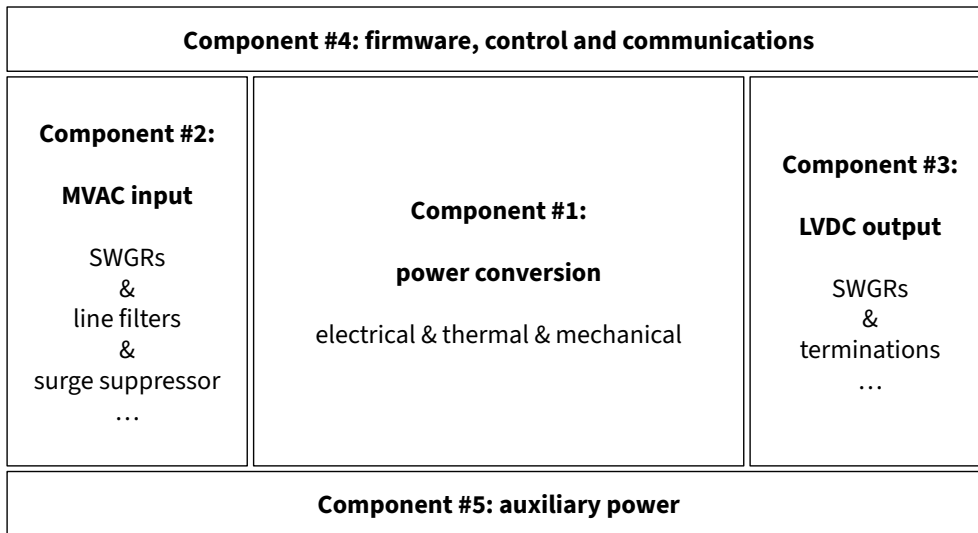


Figure 1. Main compositions of SST product (not proportional for physical size)

Component #1: Power conversion. Serving as the core engine of the solid state transformer, this component shall be responsible for delivering below key MVAC-LVDC power conversion functions to meet the performance specification described in this document. It combines power electronic semiconductors, magnetic components, capacitors, PCBA, etc, all integrated within supporting thermal and mechanical envelopes.

Component #2: MVAC input. Serving as the energy gateway of the solid state transformer, this portion shall be responsible for offering protections to SST from unexpected MVAC transient event and internal fault, supporting MVAC power quality requirements for filtering, as well as enabling operational sequence for SST state machine transitions.

Component #3: LVDC output. Serving as the LVDC power distribution entrance and switchboard from the solid state transformer, this portion shall be responsible for output port configurations, terminations and switchgears for distribution design and installation / servicing requirements.

Component #4: Firmware, control and communications. Serving as the “neuron” layer of the solid state transformer, this portion shall be responsible for component coordinations, system behavior control, and the management of both internal and external operational data exchange.

Component #5: Auxiliary power. Serving as the housekeeping function of the solid state transformer, this portion shall be the first step during power on, and operate with the highest level of availability, independent of the gear's operating status. It shall be allowed with multiple sources of power - externally and internally.

4.2.SST product configurations - MVAC and power rating

Parameter	MV 13.8kV SKU A	MV 34.5kV SKU B
Input Voltage	13.8kVac +/-10%, 3 phase 3 wire + PE	34.5 kVac +/-10%, 3 phase 3 wire + PE
Input Frequency	57 - 63Hz (50/60Hz), up to 5Hz/s as ROCOF	57 - 63Hz (50/60Hz), up to 5Hz/s as ROCOF
Power Rating (DC output)	5 MW	5MW, 10 MW
Input Power Factor	> 0.99 at full loading > 0.98 @ load ≥20% > 0.95 @ load < 20%	
MVAC Harmonics	IEEE519, IEC 61800-3 (Category C4), IEC 61000-6-2 & -6-4	
MVAC Grounding System	Low resistance grounded system or Solid grounding (Utility design based)	
Short-Circuit Fault Current Capability (kA)	18-40kA symmetrical RMS for TBD seconds (driven by appropriate industrial standard practice); ~104kA for asymmetrical value	25-50kA symmetrical RMS for TBD seconds (driven by appropriate industrial standard practice); ~130kA for asymmetrical value
Basic Impulse Level (BIL)	>= 110 kV BIL @ 13.8kV	>= 150-200 kV BIL @ 34.5 kV
Recommended MVAC Load Acceptance and Rejection	Approx. 20~50MW/min for utility; TBD MW/s for gensets (fossil fuel based)	Approx. 20~50MW/min for utility; Other energy sources TBD (renewables)

4.3. SST product configurations - LVDC definitions

Output ports	1, as illustrated in example reference LVDC architectures here (future product specification shall finetune based on application use case scenarios)
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Output Voltage (nominal)	800Vdc unipolar system +/- 0.5% (switching frequency ripples) +/- 1% (static tolerance, 0-100% loading) +/- 3% (dynamic regulation, load step TBD)
DC grounding	IT ground - floating or high resistance ground (HRG) with detection and protection coordination throughout the LVDC distribution architecture, detailed design TBD
Overload Rating (recommended setting)	- Fault protection coordination: 150% under constant current mode for 150 milliseconds, SST allowed to trip after the 150ms - Overload capacity: constant voltage mode - 120% for 5 seconds
Output DC link cap size (recommended setting)	Maximum 10mF per 4 MW output (TBD, to be finetuned after full LVDC system simulation and testing)
Efficiency	>= 98% between 50-100% utilization (power train loss only)
Dimensions (all inclusive)	Physical envelope range: 10-20' L x 8' D x 8.5' H Note: the restrictions on height requirement comes from a. transportation; b. serviceability at installation site
Auxiliary power - 1 (for control and telemetry)	Source #1: External 480Vac (backed up by data center system UPS) Source #2: internally generated
Auxiliary power - 2 (for converter semiconductor actuation and cooling)	System internally generated
Use	Option #1: indoor, grey space, environmentally conditioned Option #2: outdoor
Communications	Modbus TCP/IP

5. Electrical

5.1.Upstream MVAC interface

5.1.1.Upstream grounding and wiring configurations

In typical data center application environments, the upstream transformer medium-voltage interface utilizes a **Wye (Y) secondary winding** with a **low-resistance grounded (LRG) neutral** or **solid grounding**.

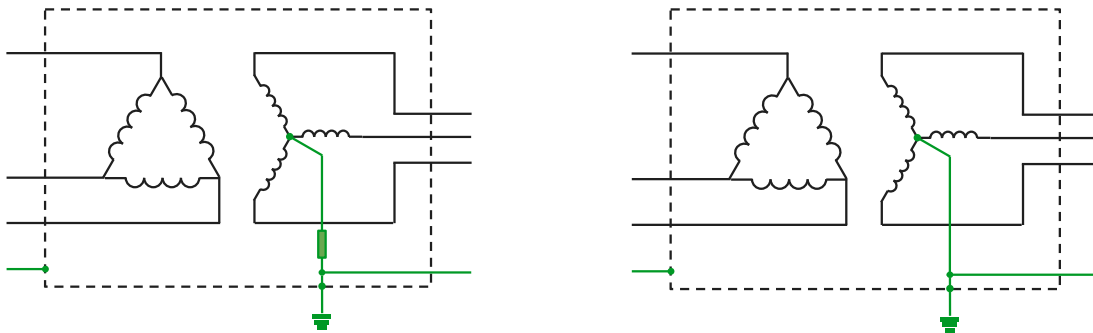


Figure 2. Upstream grounding mechanism - LRG (left) and solid grounding (right)

5.1.2.Voltage Ride Through (VRT)

Voltage ride through requirement in this section applies to conversion and energy storage systems between MVAC to LVDC - example structures such as:

- SST and ESS operating in conjunction, with ESS size dependent upon upstream energy system continuity, to a joint LVDC bus with coordinated secondary layer of control, and deliberate power sharing performance design target.
- integrated SST and ESS as a single product SKU, functioning as an “MVUPS” application.

There has not been a unified standard on the LVRT design criteria. This part of the design specification is not intended to drive a consensus, but provide guidance on the coordination between SST and ESS converter for power electronics design and control; when it comes to amount of energy required to “ride-through” the curves defined by different local utility authorities, ESS capacity can be sized differently and integrated as part of the product SKU at times of deployment (aka. product specification).

After surveying relative utility study authorities on the grid-code definitions, ERCOT [1], Fingrid, ITIC (classic PSU ride-through), IEEE1547 [2] (DER reference), IEEE2800 [3] (DER reference at transmission level), MISO, and other utility authorities, are pictured in the section below as reference. Early versions of the SST specification prioritize most recognized utility operator authorities defined LVRT definitions.

Zoom in on the definition of SST’s behavior, with ERCOT graph as illustration - grey zones in the figure below translate to SST’s behavior expectations at times of transient and permanent voltage quality conditions.

- In the **OVRT range** ($1.1\text{pu} < x < 1.2\text{pu}$), SST shall be fully functional and supply 100% of the rated output load.
- In the **acceptable power quality range** ($0.9\text{pu} < x < 1.1\text{pu}$ for ∞), SST shall be fully functional and supply 100% of the rated output load.

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- In the **LVRT ranges**:
 - In the **LVRT range #1**, SST shall be controlled to source no more than nominal input current, while supplying 100% of the rated output load. SST coupled energy storage and inherent power sharing control shall be designed accordingly to meet these performance requirements.
 - In the **LVRT range #2**, SST shall be controlled to source no input current. SST coupled energy storage shall provide 100% of the rated output load.
 - When system voltage and frequency return back to acceptable power quality immediately following any LVRT event, SST shall restore operations immediately without any state machine transitions.

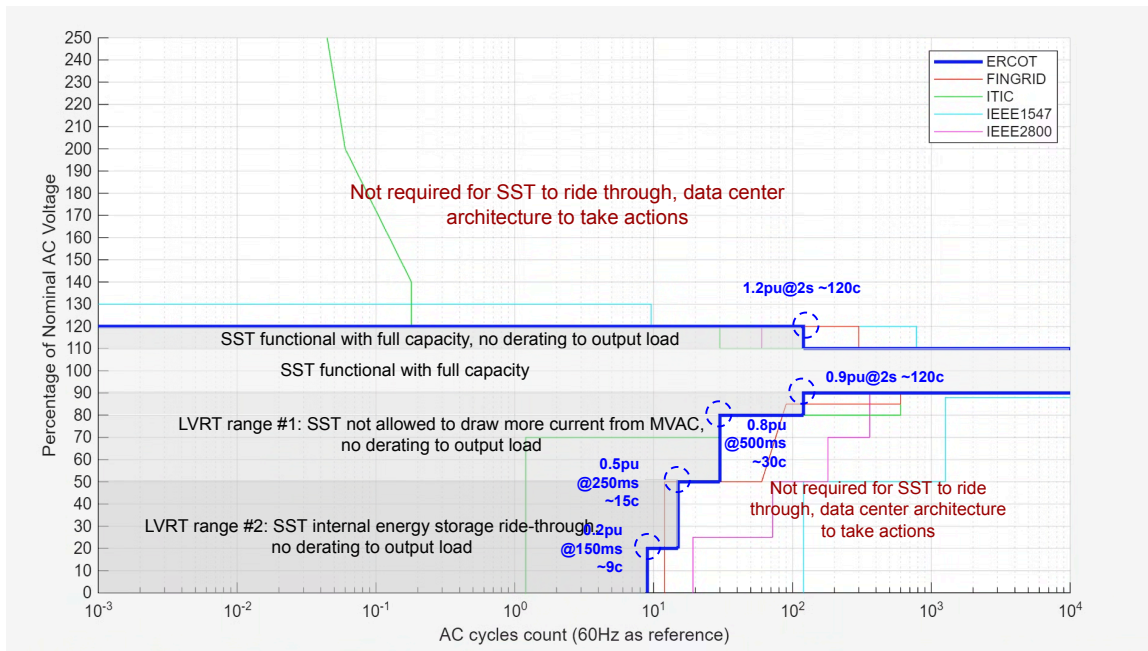
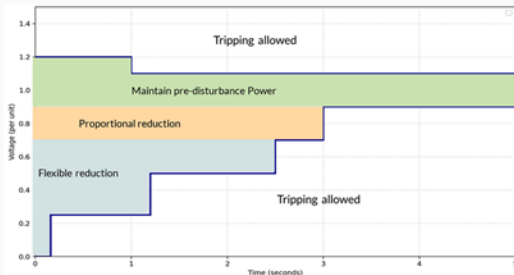


Figure 3. Voltage Ride-through requirements for SST

MISO's proposed voltage ride through requirement



0.90 – 1.20 pu

CONTINUOUS RANGE

Maintain pre-disturbance active power. Control stability is mandatory. No power reduction is permitted between 0.9 and 1.2 pu.

0.70 – 0.90 pu

MANDATORY RECOVERY

Active power reduction is permitted but must be proportional. Restoration of $\geq 90\%$ power is required within 1s of recovery.

< 0.70 pu

STABILITY PRIORITY

Flexible power reduction is allowed to maintain synchronization.

Phase Jump

Remain connected for sudden phase angle changes up to 25° .

V/Hz Limit

Ride through 1.1 pu for 45s or 1.18 pu for 2s.

Consecutive

Must handle three voltage disturbances within 10 seconds.

Transient VRT

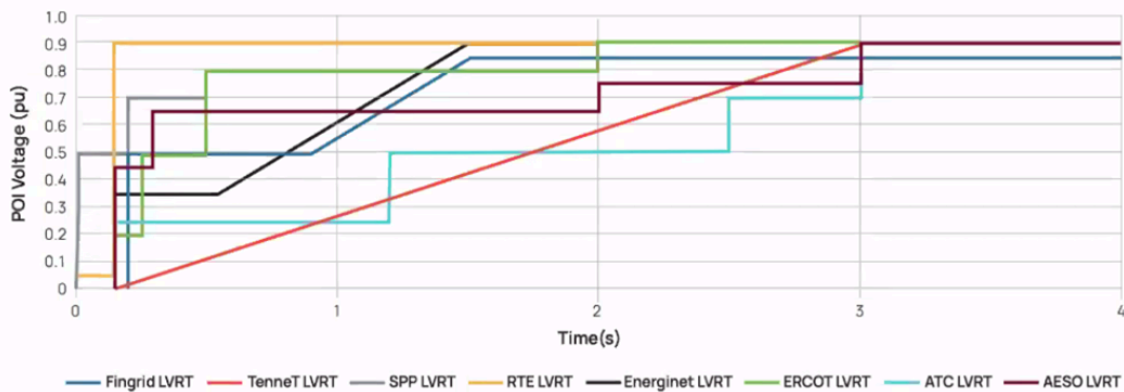
No tripping for normal system events like faults or switching

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[MISO Example]

Comparison of Voltage Ride-Through Requirements: Low VRT Profiles



Notes: AESO = Alberta Electric System Operator; ATC = American Transmission Company; ERCOT = Electric Reliability Council of Texas; LVRT = Low-voltage ride-through; RTE = Le Réseau de Transport d'Électricité; SPP = Southwest Power Pool.

Source: Energy Systems Integration Group.

5.1.3. Frequency Ride Through (FRT)

Frequency (f) (Hz)	Minimum Ride-Through Time (seconds)
$f > 61.8$	May ride-through or trip
$61.2 < f \leq 61.8$	299

$58.8 \leq f \leq 61.2$	continuous
$57.0 \leq f < 58.8$	299
$f < 57.0$	May ride-through or trip

5.2. Downstream LVDC interface

5.2.1. Voltage and power quality

- a. SST output voltage range
 - Converter's static range : 0.5% (ripples) + 1% (static regulation/static droop) = 1.5%
 - Converter's dynamic regulation range : 3% (control loop response)
- b. (TBD, pending on busway conveyance parameter) SST output to rack level DC/DC input
 - LVDC conveyance parasitics : 4% (Ldi/dt)
- c. (TBD, pending future definitions) voltage levels for seamless redundancy design
 - Block redundancy design: voltage delta between primary and catcher
 - Distributed redundancy design (finer granularity, power sharing design/dynamic droop)
- d. (TBD, pending future definitions) power and energy management coordinations

5.2.2. Verification

SST shall be tested with the load. Detailed profiles will be provided in future versions. SST output terminal voltage shall maintain within the voltage tolerance band described in the above section. Test cases covering emulated data center load profiles summarized in basic characterization terms.

5.3. MVAC Load Acceptance and Rejection

Multiple types of energy sources exist in the MVAC energy plane as below categories. They offer different ramp rates when it comes to load acceptance and rejection.

- utility
- fossil fuel based on-site generation
- renewable energy

Current data center infrastructure has ramp rates defined as approx. 20~50MW/min for utility, TBD MW/s for gensets (fossil fuel based) and TBD MW/s for renewable energy. These data are selected based on knowledge of existing data center energy infrastructure, which are subject to updates when new results become available. Coordination between SST and ESS shall be designed with the utility ramp rate as reference for system control design target.

5.4. Grounding

See high level spec for functional description. Details will be provided in later revisions of the specification.

5.4.1. Grounding impedance

5.4.2. Insulation monitoring and protection

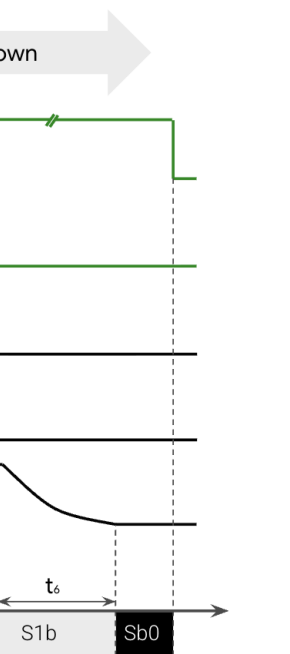
6. Environmental and structural

Parameter	Requirement
Equipment Rating	NEMA 3R Outdoor rated
Design Life	15+ Years
Environmental conditions	-30°C ~ 40°C The Supplier to provide a derating factor above 40°C
Corrosion Class	C5
Relative Humidity	95%, no condensing
Site Elevation	≤ 1000 m The Supplier to provide a derating factor above 1000m
US Land Transportation Design	per DOT FMCSA
Forward Deceleration	0.8g
Rearward Acceleration	0.5g
Lateral Acceleration	0.5g
Risk Category	II
Seismic Design Category	D
SDS	1.193g
Uniform Ground Snow Load	48 psf
Ultimate Wind Speed	120 mph (3-Second Gust) Exposure Class C

7. System behavior

7.1.State machine and transitions

Abbr.	Status	Description
S0	OFF	SST shall be in this state when “freshly getting off the dock”.
Sb	Standby	Auxiliary power on, and control system alive with visibilities into the status for the gear. The system shall park in this state, when: a) SST provisioning process is done and no fault discovered. b) SST getting ready to be powered down. Whichever scenario, SST is ready to proceed to the next state.
Sp	Precharged	This is an interim status for SST. It means internal energy storage is charged, and ready to convert power with little to none turnaround time. The source of precharge can come from either MVAC precharge circuit or ESS for self start.
S1	ON	SST is healthy. It is in steady state operations and converting power. SST’s energy source is MVAC.
S2	FAULT	SST is not healthy. It stops converting power, with aux control power staying on. Compared with a similar state of Sb, the system is parked with fault messages. System control logic shall prevent it from powering back on.
S3	WARNING	SST is outputting power, but operating under one of the below states: A. Running on ESS - SST transitions back to S1 when MVAC recovers. B. It loses redundancy, or operates in a marginal/derated state. It still converts power, but with warning messages. a) If the failure category is permanent (such as module hardware failure, or key component approaching degradation threshold but not yet failed), system control logic shall prevent the state machine from transitioning back to S1. Such failure shall be manually validated, serviced on, and issued with reset with data center operator’s intervention, and SST shall go through a new round of provisioning before putting back to operations. b) If the failure category is temporary (such as overtemperature warning OTW, or overcurrent warning OCW), system control logic shall allow state machine transition back to S1, when such conditions and system recovered back to acceptable operating conditions. SST control system shall record the warning condition log, ready for data center operations automation hub to scrutinize. c) SST can stay in this state for a brief period of time during overcurrent protection, operating with constant power characteristics for transient response. SST shall transition back to S1 when the downstream fault gets cleared.



ate machine transitions

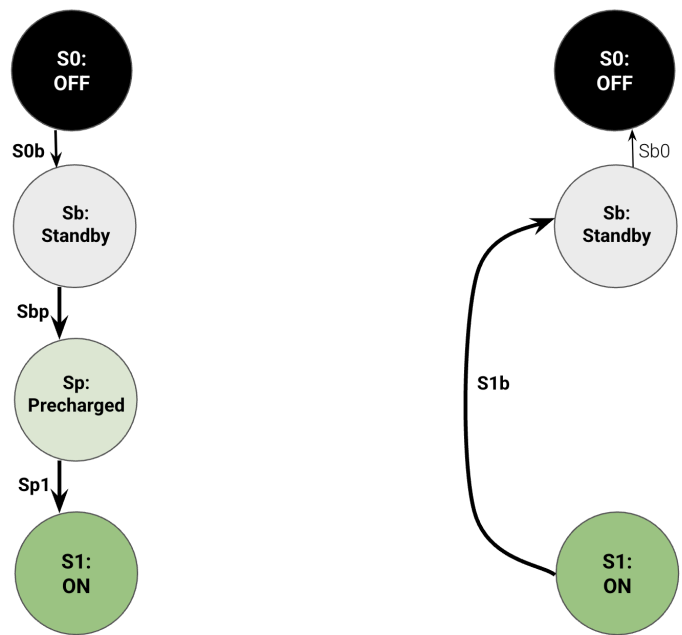


Figure 6. SST state machine transition scenarios of startup (left) and shutdown (right) without failures

8. Compliance

Additional related evolving compliance standard [source](#)

Applicable Standards & Code	
NFPA 70	National Electrical Code
NFPA 70E	Standard for Electrical Safety in the Workplace
NFPA 72	National Fire Alarm and Signaling Codes
UL 347A	Medium Voltage Power Conversion Equipment
UL 508C	Power Conversion Equipment
UL 840	Insulation Coordination Including Clearances and Creepage Distances for Electrical Equipment
UL 1741	Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources

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Applicable Standards & Code	
UL 2877	UL LLC Outline for Investigation for Power Supplies, Medium Voltage
UL 60730	Automatic Electrical Controls
UL 1699	Standard for Arc-Fault Circuit-Interrupters
IEEE 693	IEEE Recommended Practice for Seismic Design of Substations
IEEE 1584	IEEE Guide for Performing Arc-Flash Hazard Calculations
IEEE 1547 Series	Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Power System Interfaces
IEEE 2030.2	Guide for the Interoperability of Energy Storage Systems Integrated with the Electric Power Infrastructure
IEEE 519	IEEE Recommended Practices and Requirements for Harmonic Control in Electric Power Systems
IEEE 1662	IEEE Recommended Practice for Design and Application of Power Electronics in Electrical Power Systems
IEEE 1676	IEEE Guide for Control Architecture for High Power Electronics (1MW and Greater) Used in Electric Power Transmission and Distribution Systems
IEEE 1100	IEEE Recommended Practices for Powering and Grounding Electronic Equipment
IEEE 1653.6	IEEE Recommended Practices for Grounding of DC Equipment Enclosure in Traction Power Distribution Facilities
IEEE 3007.3	IEEE Recommended Practices for Electrical Safety in Industrial and Commercial Power Systems

References

- [1] ERCOT NOGRR282, latest update Jan 7th, 2026.
- [2] "IEEE Application Guide for IEEE Std 1547™-2018, IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces," in *IEEE Std 1547.2-2023 (Revision of IEEE Std 1547.2-2008)* , vol., no., pp.1-291, 20 May 2024, doi: 10.1109/IEEESTD.2024.10534228.
- [3] "IEEE Standard for Interconnection and Interoperability of Inverter-Based Resources (IBRs) Interconnecting with Associated Transmission Electric Power Systems," in *IEEE Std 2800-2022* , vol., no., pp.1-180, 22 April 2022, doi: 10.1109/IEEESTD.2022.9762253.
- [4] Additional relevant reference on codes and standards from Open Compute Project Data Center Facilities LVDC Power Distribution compliance workstream:
[+ OCP LVDC Power Distribution_Codes & Standards Master Inventory](#)

Appendix A - Example DCH distribution architecture

Below is the illustration of a block redundancy based LVDC architecture, in which the converter product described in this specification could be applied towards. The one in the dotted frame is covered by the scope for the specification.

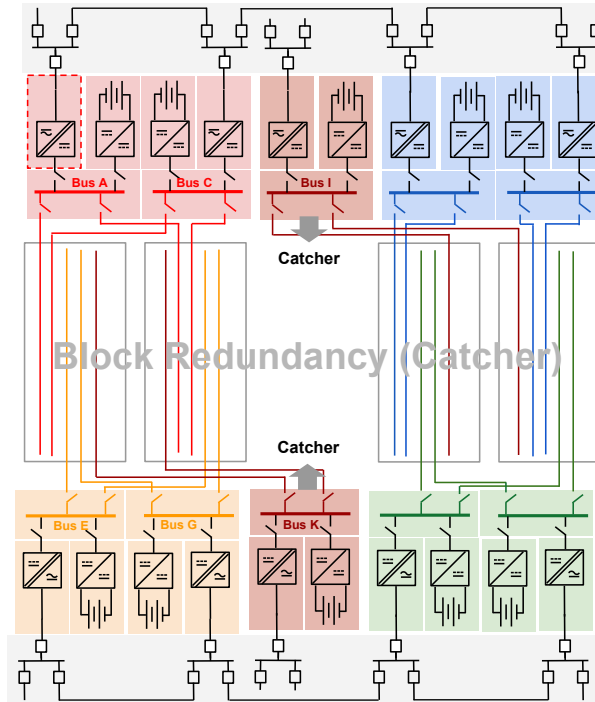


Figure 7. Example of distribution redundancy system scheme of block redundancy (catcher)

Below are other examples of SST integration for reference.

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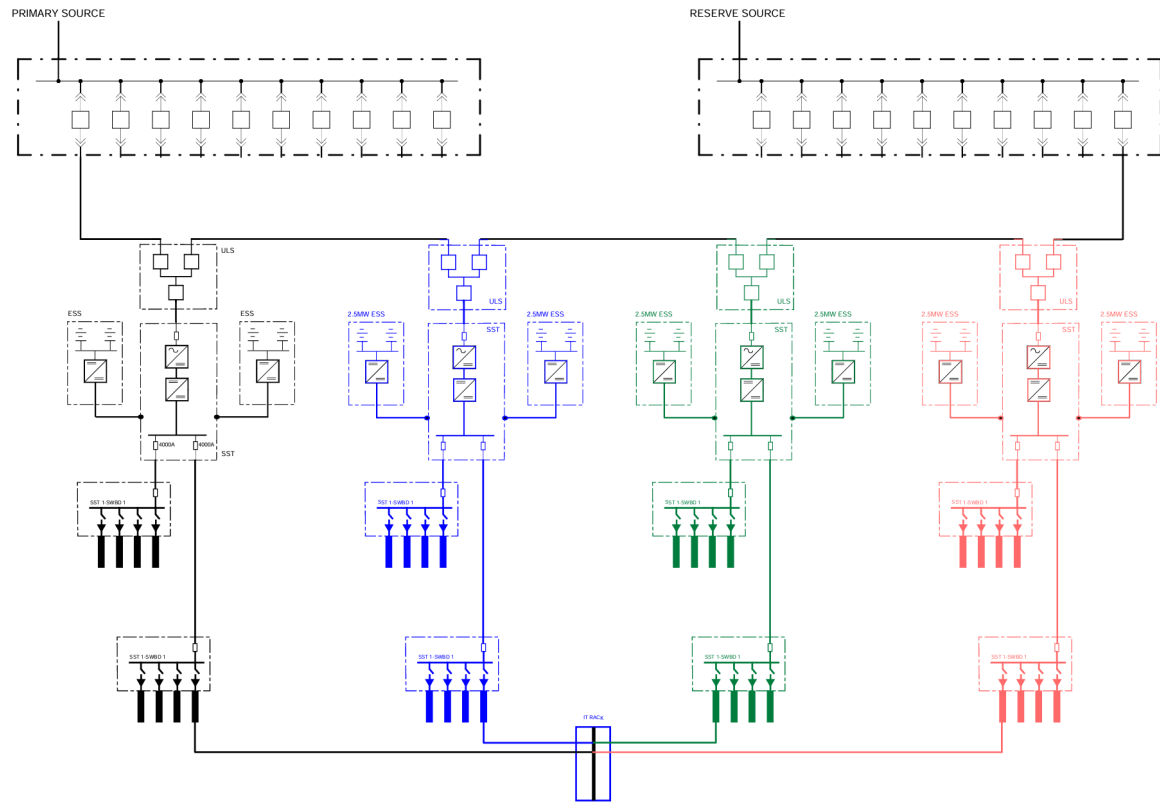


Figure 8. Example of SST Integration Busway (distributed redundant)

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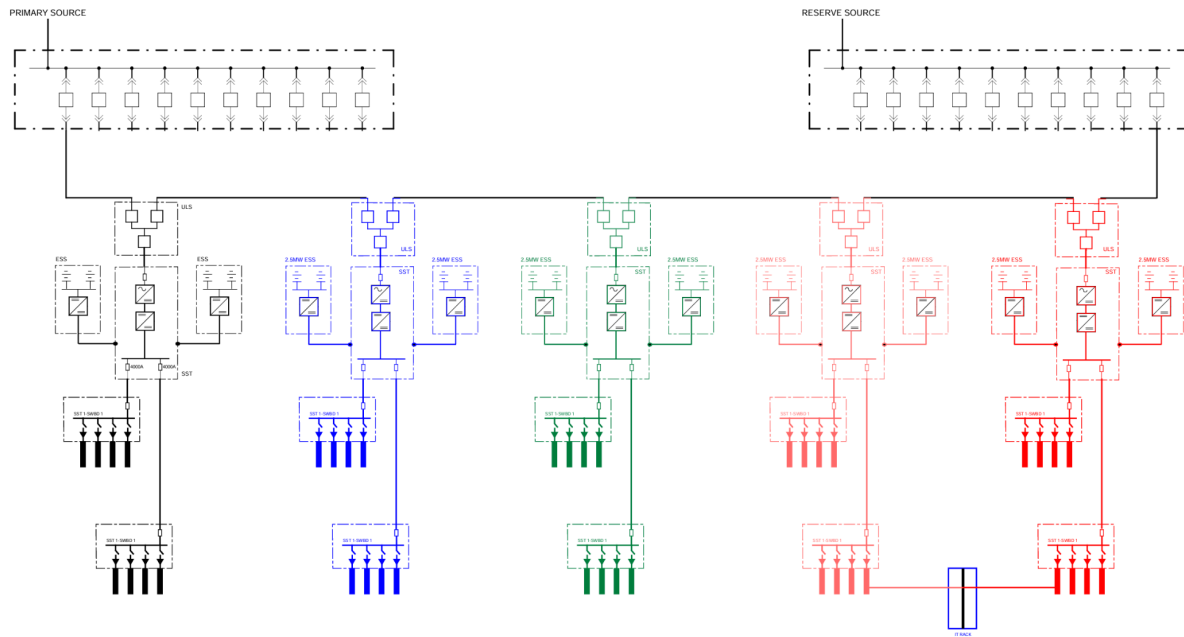


Figure 9. Example of SST Integration-Busway (N+C block redundant)

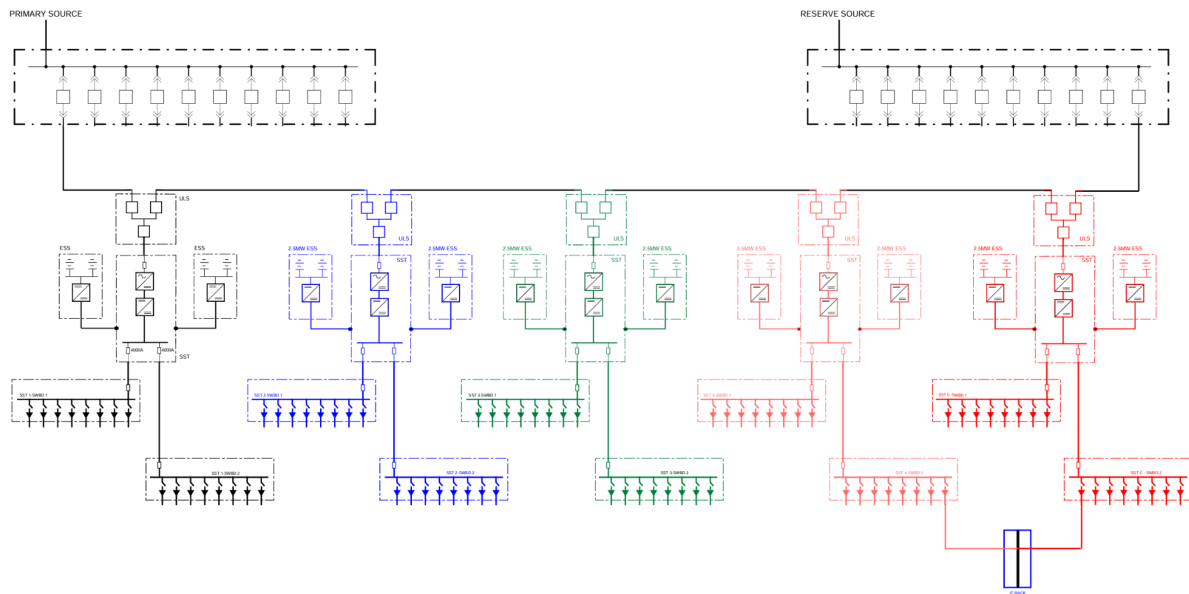


Figure 10. Example of SST Integration-Switchboard (N+C block redundant)

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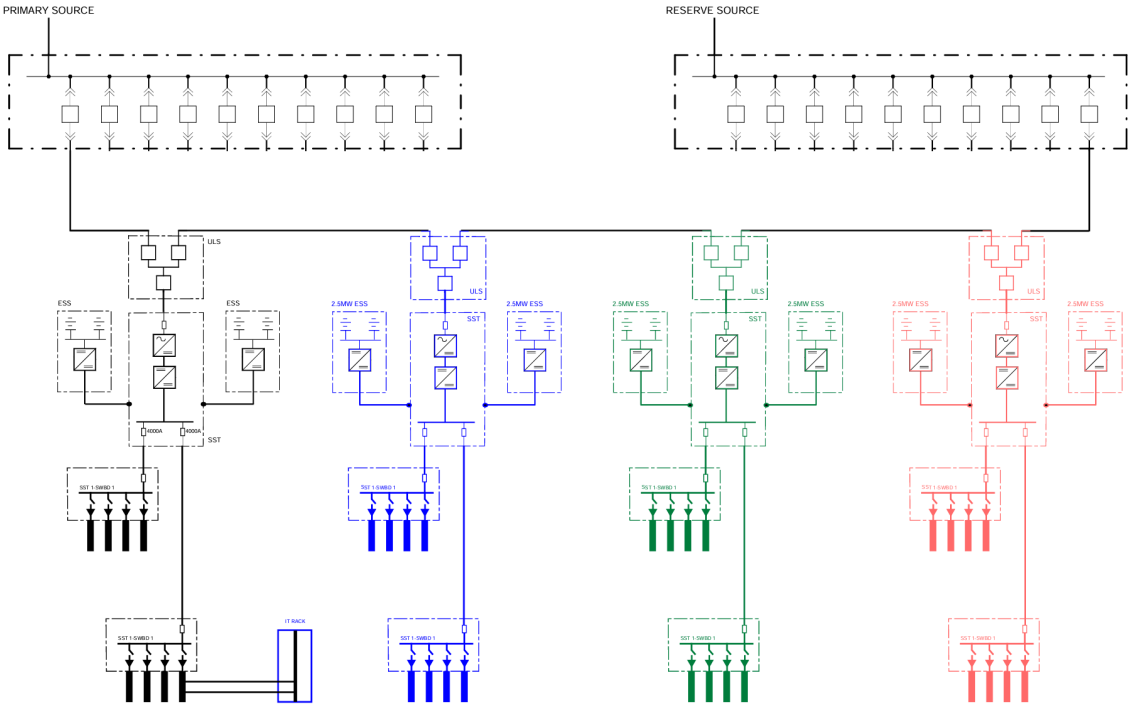


Figure 11. Example of SST Integration-Busway (N redundant)